Ramu Roy

Embedded Systems Engineer

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Summary

Embedded systems engineer working across the whole stack — PCB design in **KiCad**, ESP32/STM32 firmware, custom **Yocto** Linux and on-device ML in **Rust**. At Elipse I design the control boards for the eOS fleet and work on its Linux distribution. The UWB crane anti-collision system I built at Radiogeet, hardware and firmware, is deployed at **Tata Steel Colors**.

Work Experience

Embedded Systems Engineer, Elipse May 2026 – Present

- **Hardware / PCB design**
- Design the fleet's control boards in KiCad end to end — schematic capture, layout, design-rule checks and fabrication release — across five boards. Currently designing a Bluetooth LE remote (ESP32-H2) with a charging dock.
- **Room Controller:** re-spun from v1 to v2 at **57 % less board area**, with eight dimmable 24 V PWM channels, addressable RGB outputs and a five-stage input protection chain (fuses, eFuses, TVS). Driving the FET gates directly from the PWM expander removed four gate-driver ICs. Deployed and in service.
- **Zone Controller:** 48 V, sixteen 12-bit PWM channels, 24 A / 1,152 W, on a four-layer impedance-controlled stackup with 100 Ω differential Ethernet pairs; the ESP32 is the RMII clock master, which removed a clock buffer and four support parts. Also designed its two-layer successor with an isolated RS-485 Modbus RTU link and 14 outputs from 1 A to 4 A, checked with DC-drop, thermal and short-circuit analysis.
- **Room-sensing node:** ESP32-S3 Modbus slave on the RS-485 bus with ports for temperature, humidity and light sensors, mmWave radar, a microphone array and IR blasters; layout generated with Python scripting in KiCad. **Switchboard:** SELV wall keypad with an ESP32-S3, 30 addressable indicators and a CAN uplink.
- Bench bring-up and rework: power-path debugging, fault isolation and repair.
- **eOS — custom Yocto Linux distribution for Raspberry Pi 5**
- Author Yocto recipes for system services, ML model assets and access-control policies, with IPK packaging for incremental on-device updates; own the deployment flow from cross-compilation and WIC images to RAUC A/B OTA verification.
- Work across the Linux service stack: systemd units, D-Bus interfaces, a Mosquitto MQTT broker hardened with TLS and ACLs, and SQLite schema migrations tuned for concurrent writers.
- Designed and built a Rust per-room sensor-fusion framework (thermal, mmWave radar, air quality, light, PIR) with calibration over D-Bus.
- Built the on-device voice subsystem: a transfer-learned wake-word detector (PyTorch → ONNX → tract, pure Rust), multi-microphone best-source fusion across ESP32 satellites, Whisper STT and Piper TTS with barge-in.
- ESP-IDF firmware for ESP32 sensor satellites: BLE provisioning with on-chip EC P-256 keys and X.509 certificate exchange with the hub, NVS lifecycle across OTA updates, and time-synced microphone audio streaming.

Freelance PCB Design (Remote) Oct 2026 – Present

- Corrected the schematic and completed placement and routing of a client's secure USB-stick board

(STM32U5, eMMC, secure element), delivering the full design files with a 55-page design guide.
– Redesigning a client's 1S power bank: new charger, output stage, fuel gauge and safety microcontroller, with a dedicated two-layer pack BMS board.

Embedded Engineer, Radiogeet Sep 2025 – Mar 2026

– Built the hardware and firmware for an industrial UWB crane **Anti-Collision System** deployed at **Tata Steel Colors** (formerly Tata BlueScope Steel): real-time proximity detection and zone-based safety logic that stop hazardous crane movements.
– Chose the system's hardware, wired and integrated the Masibus DO cards and the 8-channel industrial relay system, built the RS485 links between nodes, and tested and commissioned the system on site.
– Dual-core ESP32-S3 firmware: one core for time-critical UWB ranging, the other for zone logic and an embedded web interface; ESP-NOW links between nodes, and relay control over MODBUS RTU / RS485.
– Interfaced sensors and ADCs with STM32 (timers, ADC, DAC in STM32CubeIDE); worked with RS485, LoRa and Masibus industrial I/O cards.

Research and Development Engineer Intern, Ampnics Mar 2025 – Sep 2025 (Hybrid)

– Designed and reviewed PCB schematics and layouts for open-source hardware, and supported rapid prototyping through circuit testing and debugging.

Projects

Ember — personal OS with a hand-written Rust init and shell (2026)
A Linux kernel and Yocto userland with systemd replaced by two programs written from scratch in Rust: `spark`, a PID 1 and service manager, and `hearth`, the login shell. Boots real laptop hardware over UEFI with signed A/B updates and automatic rollback.

Skills

Languages	C, C++, Rust (async / Tokio), Python, MATLAB
PCB & Hardware	KiCad schematic and layout, impedance-controlled stackups, differential-pair routing, power and protection design, fabrication release, BOM & sourcing, bring-up and rework
Embedded Linux	Yocto / BitBake / OpenEmbedded, kas, RAUC A/B OTA, UEFI boot, BSP and device tree, systemd, D-Bus, MQTT (TLS, ACL), SQLite
Firmware	ESP-IDF (ESP32, ESP32-S3), FreeRTOS, STM32 HAL/LL, BLE, OTA; UART, SPI, I^2C, CAN, RS485 / MODBUS RTU, ESP-NOW, LoRa, UWB
On-device ML	PyTorch, ONNX, tract, keyword spotting, Whisper STT, Piper TTS
Tools	Git, Qt6/QML, Python scripting for KiCad

Education

2022 – 2026 B.Tech, Electronics and Communication Engineering, **RGUKT Srikakulam** CGPA 8.3/10
2020 – 2022 Pre-University Course, **RGUKT Srikakulam** GPA 9.63/10

Certifications & Languages

2025 **NPTEL: Embedded Sensing, Actuation and Interfacing Systems** — certificate (85%)
2025 **NPTEL: Electronic Systems Design — Circuits & PCB Design** — certificate (88%)
Languages English (Proficient), Telugu (Native), Hindi (Native), Japanese (Professional working)